

DOMINIC MARCHESE

Research Assistant, Federal Reserve Bank of Atlanta

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RESEARCH INTERESTS

Empiric Industrial Organization • Econometrics • Computation Economics • Applied Microeconomics • Computational Methods

EDUCATION

Drexel University, Philadelphia, PA

Bachelor of Science in Economics and Mathematics

March 2026

GPA: 3.8

Dean's List (2023–2026)

Omicron Delta Epsilon, International Honor Society in Economics

Laurel Ridge Community College, Stephens City, VA

Associate of Arts in Business Administration

June 2023

GPA: 3.9/4.0

Dean's List (2021–2023)

RESEARCH EXPERIENCE

Federal Reserve Bank of Atlanta

Research Assistant

June 2026 – Present

- Conduct economic research and empirical analysis under the supervision of Federal Reserve economists.
- Apply econometric, statistical, and computational methods to policy-relevant economic questions.

C. & J. Nyheim Plasma Institute

Research Assistant

March 2025 – April 2026

Research Areas: Statistical Mechanics, Particle Simulation, Computational Modeling

- Developed a data pipeline for approximately 10 TB of ELFIN satellite observations using Python and SQL.
- Reduced storage requirements by roughly 70% through automated preprocessing and cleaning procedures.
- Implemented balanced sampling methods reducing computational costs by approximately 90%.
- Constructed large-scale numerical simulations in Julia involving over one million particles.
- Investigated superdiffusive and subdiffusive particle transport in Earth's magnetosphere using stochastic methods.

Drexel University School of Economics

Research Assistant

September 2025 – April 2026

Advisor: Christopher Laincz

Research Topic:

Firm Dynamics and Endogenous Growth

- Conduct computational analysis of endogenous growth models incorporating firm age dynamics.
- Investigate potential explanations for recent aggregate productivity slowdowns and barriers to firm expansion.

RESEARCH PROJECTS

Electron Super- and Sub-Diffusion in Velocity Space Due to Nonlinear Resonance with Whistler-Mode Waves

Federal Reserve College Challenge - Lead Presenter

TEACHING EXPERIENCE

Drexel University

Tutor, Microeconomics and Macroeconomics

December 2024 – April 2025

- Tutored undergraduate students in introductory economics courses.
- Improved student performance through individualized instruction and exam preparation.
- Adapted teaching methods to varying student backgrounds and learning styles.

TECHNICAL SKILLS

Programming Languages:

Python, Julia, R, Stata, SQL, LaTeX

Methods:

Econometrics, Causal Inference, Panel Data Analysis, Monte Carlo Methods, Numerical Simulation

HONORS AND AWARDS

- Dean's List, Drexel University
- Dean's List, Laurel Ridge Community College
- Omicron Delta Epsilon, International Honor Society in Economics
- Eagle Scout, Boy Scouts of America

PROFESSIONAL MEMBERSHIPS

- Omicron Delta Epsilon
- Drexel Economics Society